

SUKESH BIRADAR

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SUMMARY

Backend Software Engineer specializing in AI-powered systems, cloud infrastructures, and scalable REST APIs. Built production-grade Hybrid RAG pipeline integrating Azure OpenAI, vector databases, and caching for intelligent log analysis. Expertise in system design, cost-aware AI architectures, and end-to-end Azure deployment. Currently architecting intelligent observability systems using embeddings, vector search, and LLM fallback mechanisms.

TECHNICAL SKILLS

Languages: Java, Python

Backend: Spring Boot 3.3, Spring Security, Spring Data JPA, REST APIs, JWT

AI/ML: RAG (Retrieval-Augmented Generation), Embeddings, Vector Search, LLM integration, Prompt Engineering

Cloud: Microsoft Azure (OpenAI, AI Search, App Service, Cache for Redis, Container Registry, App Services)

Database: MySQL, Azure AI Search (HNSW), Redis

DevOps: Docker, GitHub Actions, Maven, Swagger

PROJECTS

AI Log Debugger – Hybrid RAG pipeline for Intelligent Log Analysis

[GitHub](#) + [Azure](#)

Java 17, Spring Boot 3.3.2, Azure OpenAI (GPT-4o-mini), Azure AI Search (HNSW), Redis, Swagger

- Engineered 9-step cost-optimized RAG pipeline reducing LLM calls by 70% through cache-first, rule-based retrieval.
- Integrated Azure OpenAI embeddings (text-embedding-ada-002) + GPT-4o-mini with dynamic fallback routing for unknown patterns.
- Designed vector store using Azure AI Search (HNSW algorithm) with auto-index creation, <50ms similarity search
- Built rule engine with 10+ semantic patterns (NullPointerException, DB failures, OutOfMemoryError, JWT issues).
- Implemented Redis caching (SSL, 6hr TTL) with SHA-256 hashing, reducing repeated analysis by 85%.
- Created multi-input sanitization handling raw stack traces, console dumps, special characters – production-grade validation.
- Deployed to Azure App Service (Linux, Java 17); Swagger UI for API documentation
- Architecture: Input Validation → Normalization → Cache → Rule Engine → Embeddings → Vector Search → LLM

ShopNexa — Cloud-Deployed E-Commerce Backend

[GitHub](#)

Java, Spring Boot 2.7, Spring Security, JWT, MySQL, REST APIs, Docker, Azure App Service, CI / CD

- Built 20+ production REST APIs (authentication, cart, orders, payments) supporting concurrent transactions
- Implemented normalized relational schema (8+ tables) with indexes – 35% query performance improvements
- Containerized with Docker, deployed via Azure Container Registry + App Service with GitHub Actions (60% faster)
- Debugged 10+ production issues: SSL DB connectivity, Schema validation, transaction rollbacks, env config.
- Tested all endpoints with Postman; handled edge cases in inventory and concurrent order processing.

EXPERIENCE

Software Developer Intern — KodNest Technology, Bangalore

Apr 2025 – Dec 2025

- Developed backend services across 3+ modules using Spring Boot layered architecture.
- Optimized 20+ SQL queries; resolved 12+ defects improving stability and runtime errors.
- Built 15+ REST API endpoints with validation, error handling, transaction management
- Collaborated in code reviews, contributing maintainable, production-ready code.

EDUCATION

Bachelor of Engineering — Computer Science

2021 – 2025

Bheemanna Khandre Institute of Technology, VTU | CGPA: 7.55 / 10.0

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900) - [Credential](#)
- Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103) – [In Progress](#)
- Microsoft Certified: Azure AI Cloud Developer Associate (AI-200) – [In Progress](#)